AI Adoption Playbook for Regulated Infrastructure

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Executive summary

An AI use case is adopted when people do their work differently because of it and the organization can show what that change is worth. Deployment is only the halfway point. This playbook covers the whole path, from a raw idea to a measured benefit or a deliberate stop, for organizations that run regulated infrastructure.

It is written for the people who carry a use case through that path: transformation and change leads, business analysts, product owners, and the business sponsors who fund the work. It assumes a governance function exists. The companion paper, *Practical AI Governance for Regulated Infrastructure*, covers risk tiering, evaluation gates and decision rights. This playbook covers everything around them: selection, preparation, people, and value.

Three ideas run through it.

1. **Adoption is designed, not announced.** Workflow change, role impacts and the reasons people will or won't trust the tool are planned before the build starts, not after go-live.
2. **Usage is not value.** Logins and queries are leading signals. The benefit case is closed only by a measured change in the outcome the use case was approved for, compared against a baseline taken before the pilot.
3. **Every stage ends in a decision someone owns.** Proceed, defer, change, scale or stop. Stopping a use case that isn't delivering is a success of the process, not a failure of the team.

The worked example is NorthRiver Energy Distribution, a fictional mid-sized Ontario distributor, taking its first use case (a governed maintenance knowledge assistant) through the six stages.

Why adoption stalls in regulated infrastructure

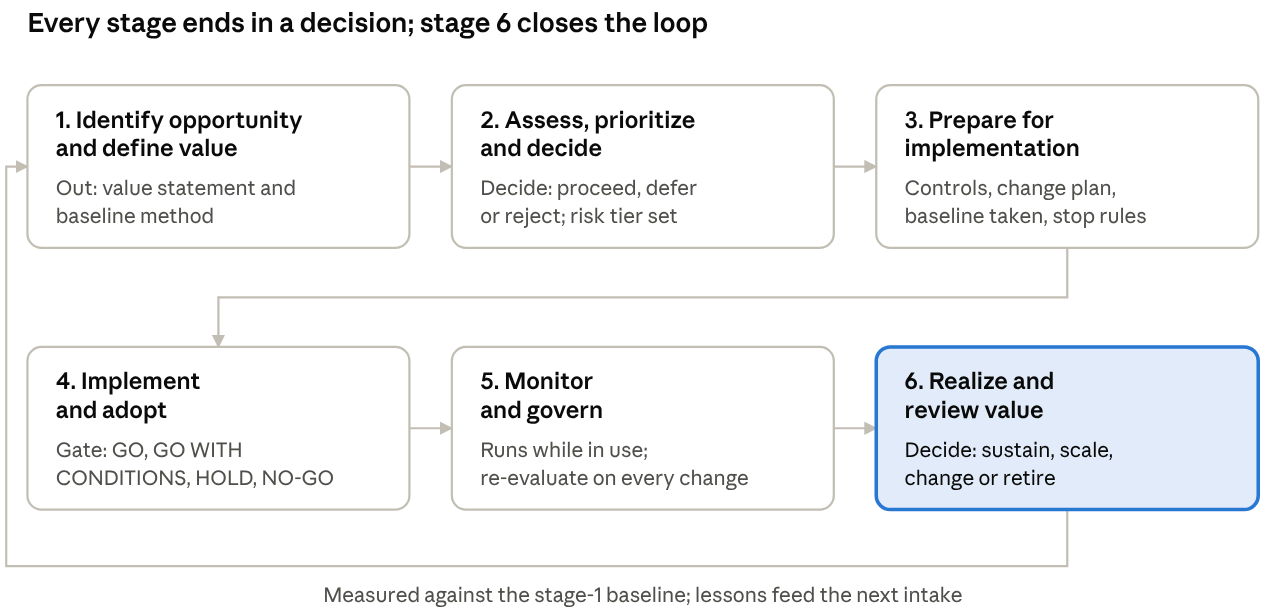
Most stalled AI programs in utilities fail for organizational reasons, not technical ones. Five patterns recur.

Pattern	What it looks like	What prevents it
Pilot purgatory	A pilot runs, users like it, and nobody decides whether to scale or stop it. It lingers on a temporary budget	Exit criteria and a named decision-maker set before the pilot starts
Usage mistaken for value	Success is reported as active users and query counts. Nobody can say whether planning got faster or errors fell	A value chain from usage to outcome, with a baseline taken before the pilot (see *Measuring value*)
Change treated as training	A one-hour session at go-live, then surprise when usage drops after week three	A change plan built from the impact on each role, started at stage 3
Trust lost on one bad answer	A field crew sees one confident wrong answer and stops using the tool. The story spreads faster than any metric	Grounded-or-refuse design, visible citations, and a fast route to report a bad answer
Labour relations raised late	The bargaining unit hears about the tool from members, assumes monitoring or job loss, and opposes it	Early, specific engagement on what the tool does and does not do, written into the change plan

Regulated infrastructure makes each of these sharper. Work is safety-critical, so tolerance for wrong answers is low. Workforces are often unionized and long-tenured. Regulators and auditors expect decisions to be traceable. And benefits arrive as staff time and risk reduction, which are harder to show than revenue.

The adoption path

A use case moves through six stages, and each one ends in a recorded decision with a named owner. The stages describe the state the use case must reach, not which team does the work, so they survive a reorganization.



adoption path · 6 stages, each closed by a decision

Risk is assessed once at stage 2 and watched continuously at stage 5. Value is defined at stage 1 and only proven at stage 6, against the same baseline. Stage 4 holds two things that fail independently: a solution can be fully deployed and still not adopted.

Plays by stage

Each play lists the few moves that matter at that stage and the decision that closes it. Templates are named in brackets; they are kept deliberately short.

Stage 1: Identify opportunity and define value

1. Write the problem without naming a technology. "Planners can't confirm the applicable standard quickly" is a problem; "we need a chatbot" is not.
2. Name the beneficiary and the person who will be accountable for the benefit.
3. Write a value statement that can be measured: which outcome, in which unit, moving from what to what.
4. Decide how the baseline will be taken before any build starts. If it can't be measured, say so now.

Closes when: problem, beneficiary, value statement and baseline method are written down [intake form].

Stage 2: Assess, prioritize and decide

1. Score value and feasibility on the same criteria for every candidate, so they can be compared [prioritisation scorecard].
2. Assign the risk tier (governance paper, *Risk tiering*). The tier sets the controls, evaluation and approval the use case will need.
3. Compare non-AI options honestly, including doing nothing, better search, and an off-the-shelf copilot.
4. Check the result is robust: would a reasonable change to the weights change the winner?

Closes when: a named decision-maker records proceed, defer or reject, with the tier and the reasons [decision record].

Stage 3: Prepare for implementation

1. Map the current and target workflow for each affected role, step by step. Mark where the AI output is used and who checks it.
2. Build the change plan from those role impacts (see *People and change*).
3. Turn the tier's requirements into named controls and evaluation tests, each with an owner.
4. Run the baseline measurement now, with the pilot group, before they see the tool.
5. Set the pilot boundary and the stop criteria in writing.

Closes when: controls, change plan, benefit owner, baseline and pilot boundary are agreed. A discovery gate here can still send the use case back if the data or content isn't fit.

Stage 4: Implement and adopt

1. Build or configure in small slices, each one evaluated before the next.
2. Hold the evaluation gate: the decision is GO, GO WITH CONDITIONS, HOLD or NO-GO, never an average score.
3. Launch to the pilot group with champions in place and a one-click way to report a bad answer.
4. Track adoption weekly. Treat a fall after week three as a signal to investigate, not a reason to send reminders.

Closes when: the solution is live within its approved scope, its controls are operating, and adoption is measured, not assumed.

Stage 5: Monitor and govern

1. Re-run the evaluation set on every material change: model version, prompt, content, or configuration.
2. Review reported bad answers and reviewer overrides together each week during the pilot, then monthly.
3. Watch for drift in how the tool is used. People find uses nobody approved, and those need a decision.

Closes when: it doesn't. This stage runs for as long as the solution is in use.

Stage 6: Realize and review value

1. Re-run the baseline measurement with the same method, group and tasks.
2. Compare against the stage-1 value statement, not against a new target set after the fact.
3. Decide: sustain, scale, change or retire. Record the lessons and feed them into the next stage-1 intake.

Closes when: the decision is recorded with its evidence [benefits review].

People and change

The change plan starts from role impacts, not from the tool. For each affected role, write down what changes in their day, what they gain, what they lose or fear losing, and who they trust. The plan follows from that.

ADKAR applied to AI adoption

ADKAR (Awareness, Desire, Knowledge, Ability, Reinforcement) works well for AI because each element has a specific AI failure mode.

Element	The AI-specific question	Typical move
Awareness	Do people understand why this is changing, and what the tool will and won't do?	Lead with the problem in their words. State plainly: it advises, people decide; it does not monitor individuals
Desire	Do they believe it will make their work better rather than watched or replaced?	Involve respected practitioners in testing. Let sceptics try to break it on real questions
Knowledge	Do they know how to ask well, read a citation, and spot when to distrust an answer?	Short task-based sessions on their own questions, not a feature tour
Ability	Can they use it under real conditions, such as on site, under time pressure, on a phone?	Supported first weeks, champions on each crew, a quick way to report a bad answer
Reinforcement	Does using it stay worthwhile after the launch attention fades?	Share fixes made from reported answers. Show the team its own before-and-after measures

Where adoption stalls depends on the role and the context, so diagnose the barrier point for each affected group rather than assuming it. Two patterns are common with AI tools. Desire stalls when staff read the tool as monitoring or replacement. Ability stalls when people are trained but, under real conditions, asking a senior colleague is still faster and safer. Champions help most with the second.

Labour relations

In a unionized workforce, engage the bargaining unit at stage 3, before the pilot group is chosen. Be specific: what data is logged, who can see it, and that logs are used to improve the tool, not to assess individuals. Put those commitments in writing and design the logging to match them. A vague assurance given early and contradicted later costs more trust than the tool can earn back.

Champions

Choose champions for credibility with peers, not enthusiasm for technology. A respected senior electrician who says "it found the Lakeshore exception I would have missed" does more than any launch email. Give champions a direct line to the product owner and act visibly on what they report.

Measuring value

Value is shown by a measured change in the outcome the use case was approved for, against a baseline taken before the pilot. Usage figures support that claim; they never replace it.

The usage-to-value chain

Measure at four levels. Each level is evidence that the one below it can happen, and each fails differently.

Level	Question	Example measures	Available from	What it can't prove
1. Usage	Are people using it?	Weekly active users in the pilot group; queries per user; return rate after week 3	Week 1	That the answers were used or helped
2. Quality of use	Are they using it well and trusting it appropriately?	Share of answers opened at the citation; reported bad answers; reviewer overrides; refusals	Week 2	That work got faster or better

Level	Question	Example measures	Available from	What it can't prove
3. Task outcome	Does the task get done faster or better?	Minutes to find and confirm the right standard on a set task; same answer for the same question across areas	Mid-pilot task test	That the organization gains anything
4. Business outcome	Does the organization gain what it approved?	Planner hours released; fewer audit findings on the wrong standard; time to answer an audit query	End of pilot, then quarterly	Nothing further; this is the claim

Read the chain from the bottom. A strong level 1 with a weak level 3 means people are using it but it isn't helping: fix the content or retrieval before promoting it. A strong level 3 with a weak level 1 means it works but isn't trusted or reachable: that is a change problem, not a technical one.

Baselines

Take the baseline in stage 3, with the pilot group, before they see the tool. Use a fixed set of real tasks (for example, 10 standard questions) and time them. Self-reported estimates of time saved are useful context but too generous to carry a business case on their own. If a measure has no baseline, say so in the benefits review rather than inventing one afterwards.

Attribution

Other things change during a pilot: a new procedure, a staffing change, a quiet month. Three methods, in order of strength:

1. **Controlled task test.** The same people do the same set tasks with and without the tool. Strongest for task outcome, cheap to run.

2. **Comparison group.** A similar group without the tool (another operating area) measured over the same period. Separates the tool's effect from everything else that changed.
3. **Before and after.** The same measure, same group, before and after. Weakest, because it credits the tool with every other change. Use it only alongside one of the others.

State which method each figure rests on. A reader who sees "40% faster (controlled task test, n = 12)" can judge the claim; a reader who sees "40% faster" cannot.

Turning time into money, honestly

Released time is capacity, not cash, unless headcount or overtime actually changes. Report it as capacity value and say what the time was used for.

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\text{Annual capacity value} = \text{hours saved per person per week} \times
\text{people} \times \text{working weeks} \times \text{loaded hourly cost}
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The break-even question is more useful than the headline figure: how many hours a week per person must the tool save to cover its cost? If that number is small and the task test clears it comfortably, the case is robust. If it depends on the most optimistic estimate, it isn't.

When to stop

Set stop rules before the pilot, alongside the success criteria. Stop or change course when:

- a critical evaluation gate fails and can't be fixed within the pilot window;
- usage falls after week 3 and interviews show people don't trust the answers;
- the task test shows no meaningful improvement over the baseline;
- the break-even saving isn't reached by the end of the pilot.

A stopped use case still produces value: a tested control set, a cleaned content library, and evidence for the next decision.

Operating model and roles

A small, standing team carries use cases through the path. Each use case borrows a business sponsor and a product owner from the area it serves. The governance paper sets who approves what; this table sets who does the work.

Role	Accountable for	Most active in stages
Business sponsor	The value statement and the final sustain, scale or stop decision	1, 2, 6
Benefit owner (often the sponsor's delegate)	Baseline, benefit measures, and the benefits review	1, 3, 6
Product owner	Scope, backlog, pilot boundary, and acting on reported bad answers	3, 4, 5
Transformation / change lead	Role-impact analysis, change plan, champions, labour-relations engagement	3, 4
Business analyst	Problem framing, current and target workflow, requirements, acceptance criteria	1, 2, 3
Solution architect and build team	Design, build, integration, evaluation harness	3, 4, 5
AI governance lead	Risk tier, control set, evaluation gates, monitoring reviews	2, 3, 4, 5
Champions	Peer support and honest feedback from the front line	4, 5

Cadence. A monthly portfolio review looks at every use case in flight: its stage, its next decision, and its level 1 to 4 measures. Weekly reviews run only for use cases in pilot. Anything without a decision date is treated as stalled.

Scale. One organization rarely needs more than one portfolio review and one shared set of templates. What changes with scale is how many use cases are in flight, not how many governance bodies there are.

NorthRiver worked example

NorthRiver's first use case is a governed assistant that gives maintenance staff the current standard for a specific asset, with citations, alongside that asset's record in the enterprise asset management (EAM) system. All figures are illustrative.

Stage	What NorthRiver did	Decision and evidence
1. Identify	Framed the problem as slow, inconsistent access to the applicable standard across two pre-merger document sets, with expert knowledge retiring	Value statement: time to find and confirm the standard falls from hours to minutes. Baseline method: timed test of 10 standard questions
2. Assess	Scored six candidates on the same value and feasibility criteria; compared an off-the-shelf copilot and a document clean-up as alternatives	Maintenance knowledge ranked first (16.7 of 25, Pilot quadrant, Medium risk tier). The lead holds unless the value weight falls below 30%
3. Prepare	Mapped planner and field workflows; set a discovery gate at week 2 to confirm the content and EAM data are fit before building	Pilot boundary: one operating area's planners and one station crew, read-only, advisory. Stop on any critical evaluation failure
4. Implement	Planned: first slice is the EAM lookup with customer data filtered out, then three standard questions end to end	Evaluation gate at week 12, then a 12-week pilot
5. Monitor	Planned: evaluation set re-run on every change; weekly review of overrides and reported answers during the pilot	Not yet started
6. Review	Planned: re-run the 10-question test with the same group	Not yet started

The adoption risk that matters most

The Lakeshore operating area still follows a regional inspection practice note that was never formally approved. The assistant is designed to disclose that conflict rather than quietly pick a side. For Lakeshore staff, the first thing the tool does may be to flag their established practice as unapproved.

That is a change problem before it is a technical one, and the practice note's status is a decision for the maintenance standards owner, not for the AI project. So the pilot runs in another operating area, and the conflict behaviour is proven in the evaluation instead.

Lakeshore supervisors are briefed early, so their crews hear how the tool will treat the practice note from their own leaders before they ever see it on screen. Keeping the hardest case out of the pilot is deliberate: it protects adoption while the standards owner settles the note, and the evaluation still tests that the tool handles the conflict correctly.

The value case in measures

Chain level	NorthRiver measure	Pilot target
Usage	Weekly active users in the pilot group	At least 70% by week 8
Quality of use	Reviewer overrides; reported bad answers	Tracked weekly; every report triaged
Task outcome	Minutes on the 10-question task test; same answer across areas	At least 40% faster; repeatability of 90% or more, 100% on critical facts
Business outcome	Planner hours released; time to answer an audit query	Measured at pilot end against the stage-1 baseline

On cost: about CAD 156,000 one-time and CAD 600 to 1,100 a month to run. Each hour saved per person per week across 40 staff is worth about CAD 184,000 a year in capacity, so break-even is about 0.85 hours per person per week. The task test target, if met, clears that comfortably. Until the test is run, the savings are planning assumptions, and evaluation results on synthetic data are never presented as realized benefit.

First 90 days

For an organization starting from scattered pilots and no common path, the first 90 days should produce one use case moving properly through the stages, not a framework document.

1. **Days 1 to 30: take stock and pick one.**
    - List every AI use case in flight or proposed, including vendor-led ones, with its sponsor and status.
    - Score them on the same criteria and pick one with clear value, contained risk and a willing sponsor.
    - Agree the six stages and the decision that closes each. One page is enough.
2. **Days 31 to 60: prepare it properly.**

- Map the affected workflows and role impacts; open the conversation with the bargaining unit.
- Write the value statement, set the stop rules, and take the baseline with the pilot group.
- Name the benefit owner and the champions.

3. **Days 61 to 90: run it and report honestly.**
   - Launch the pilot, or the evaluation that comes before it, within its written boundary.
   - Hold the first monthly portfolio review with level 1 to 4 measures for every use case listed in month one.
   - Decide what happens to the other pilots found in month one: bring them onto the path, or stop them.

The last step is the one most often skipped. Leaving old pilots running outside the path undoes the discipline the new one was meant to establish.

About this document and sources

NorthRiver Energy Distribution is a fictional organization, invented to show the playbook applied end to end. Every NorthRiver figure is illustrative and drawn from no real utility. The maintenance knowledge assistant is a prototype under construction on synthetic documents and records; no pilot has run and no benefit has been realized.

The six stages come from NorthRiver's AI adoption value stream. ADKAR is Prosci's change management model, referenced by name and applied here to AI adoption.

Companion documents

- Practical AI Governance for Regulated Infrastructure: risk tiering, evaluation gates, decision rights
- NorthRiver Business Case: Maintenance Knowledge & Records Assistant: measures, baselines, costs and pilot plan behind the worked example
- NorthRiver Use-Case Prioritisation Scorecard v1.0: the six-candidate scoring and sensitivity test behind stage 2